SI 201 Final Project Report

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Initial Goals:

1. *Are there more flights during good weather conditions or bad weather conditions?* - ***Iman***
2. *What are the top five countries of origin for travelers in the Jeffrey Epstein case data?* - ***Wale***
3. *How often is an individual mentioned in the files and confirmed to have visited the island?* - ***Ezeddin***

API/Websites planned to be used:

Epstein Investigation API
https://www.epsteininvestigation.org/developers
National Weather Service API
https://www.weather.gov/documentation/services-web-api
Wikipedia (scraped with BeautifulSoup)
https://en.wikipedia.org/wiki/Main_Page

Data planned to be gathered and calculated from database:

Epstein API
Flights data: passenger_name, flight_date, origin, destination
People: name, entity_type (person / organization), document_count

Wikipedia (BeautifulSoup scraping)
will scrape: Page: https://en.wikipedia.org/wiki/List_of_people_named_in_the_Epstein_files
Data: name, occupation, notes/description

Weather API
will collect: date, temperature, weather conditions (rain, snow, clear, etc.)
(Used to test if weather correlates with flight patterns)

Growth Rate of Flights
Formula:
growth_rate = (flights_year2 - flights_year1) / flights_year1

Goals achieved:

All three original goals were achieved. Flight data was successfully pulled and analyzed with weather conditions. Passenger country-of-birth data was achieved. Individual passenger mention counts were calculated with their frequency of appearing on target destination flights (Little Saint James, Cyril E King Airport, Palm Beach international, and Miami International).

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Data gathered

Year-over-year flight growth rate from 1991–2019 using the formula (year2 - year1) / year1 * 100
Average passenger count per flight grouped by departure weather
Departure and arrival weather breakdown by destination airport
Per-person total flight mentions vs. mentions on target destinations, with ratio percentage
Passenger country of birth percentages across all passengers

Problems we faced:

- Only storing 25 items in our database each time a file is run -> we originally had hundreds of items and rows
- Finding a second API key that would relate to our project
- In the SQL database, we needed to make sure we correctly organized all the columns and that we did not have duplicate strings of data

Calculations from database:

Year-over-year flight growth rate

```
previous_count = None
for year, count in rows:
    if previous_count is None or previous_count == 0:
        growth_pct = None  # no growth value for first year
    else:
        growth_pct = ((count - previous_count) / previous_count) * 100

    yearly_growth_list.append([int(year), int(count), growth_pct])
    previous_count = count
```

Average number of passengers per flight, grouped by departure weather type

```
weather_avg_rows = cursor.execute(
    '''
    SELECT
        wc.code AS weather_type,
        ROUND(AVG(f.passenger_count), 2) AS avg_passenger_count,
        COUNT(f.id) AS flights_n
    FROM weather_conditions wc
    LEFT JOIN flights f ON f.departure_weather_id = wc.id
    GROUP BY wc.code
    ORDER BY wc.code
    '''
).fetchall()
```

Percentage of passengers born in each country out of all passengers

```
countries = [row[0] for row in rows]
counts = [row[1] for row in rows]
total = sum(counts)
percentages = [(c / total) * 100 for c in counts]
```

Per-person target destination ratio:

```
ratio_rows = []
for name, total_count in total_mentions.items():
    target_count = target_destination_mentions.get(name, 0)
    ratio_pct = (target_count / total_count) * 100 if total_count else 0
    ratio_rows.append(
        {
            'name': name,
            'total_mentions': total_count,
            'target_destination_mentions': target_count,
            'ratio_pct': ratio_pct,
        }
    )
```

Weather percentage per destination (SQL window function):

```python
# Percentage per weather label (within one destination)
cur = conn.execute(
    """
    SELECT
        COALESCE(NULLIF(TRIM(departure_weather), ''), '(empty)') AS weather,
        ROUND(100.0 * COUNT(*) / SUM(COUNT(*)) OVER (), 2) AS percentage
    FROM flights
    WHERE LOWER(TRIM(destination)) = LOWER(TRIM(?))
    GROUP BY departure_weather
    ORDER BY percentage DESC
    """,
    ("Cyril E. King Airport, USVI",)
)
weather_rows = cur.fetchall()  # list of tuples: (weather, percentage)
weather_labels = [r[0] for r in weather_rows]
percentages = [float(r[1]) for r in weather_rows]
list(zip(weather_labels, percentages))
```

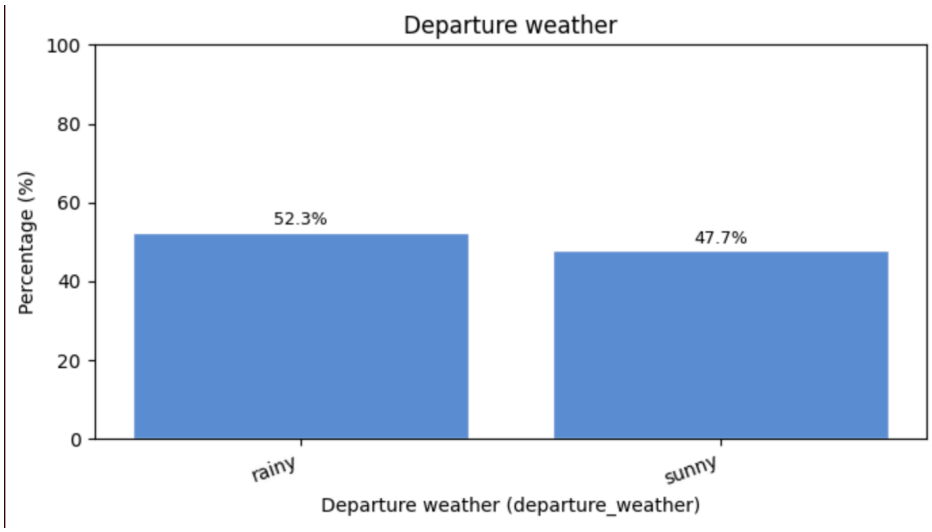
Country percentage out of all passengers:

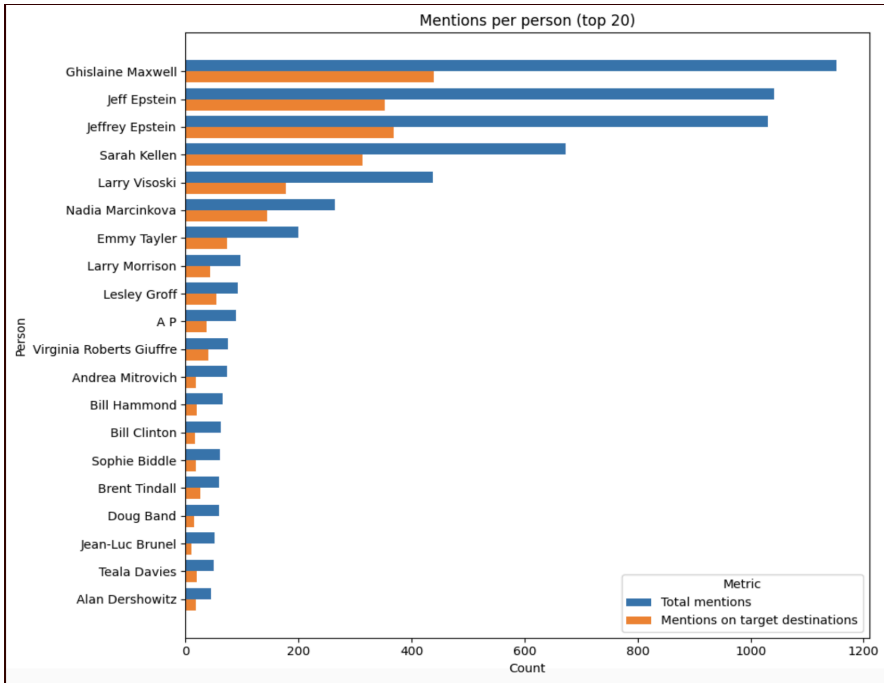
```python
countries = [row[0] for row in rows]
counts = [row[1] for row in rows]
total = sum(counts)
percentages = [(c / total) * 100 for c in counts]
```

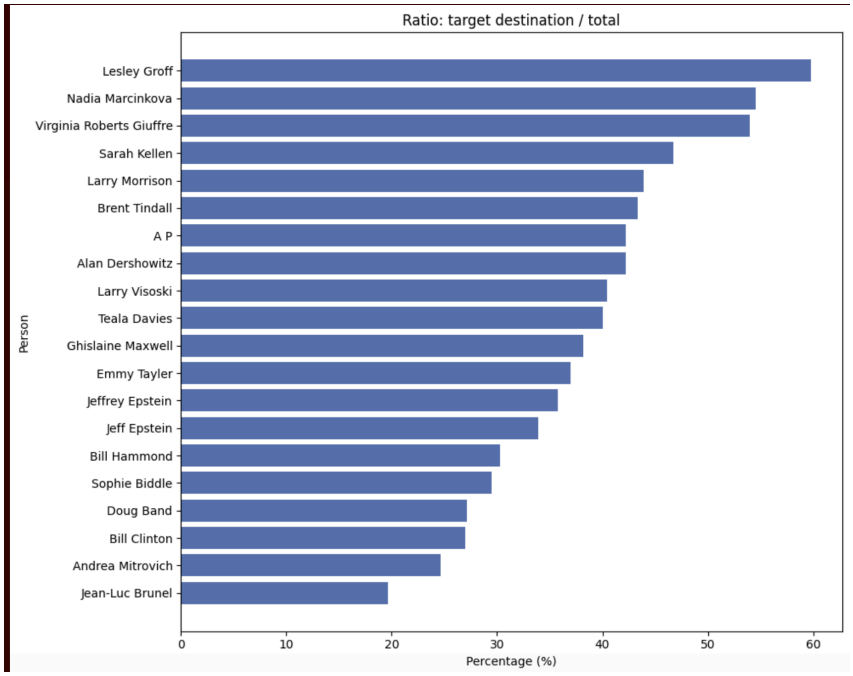
Weather classification:

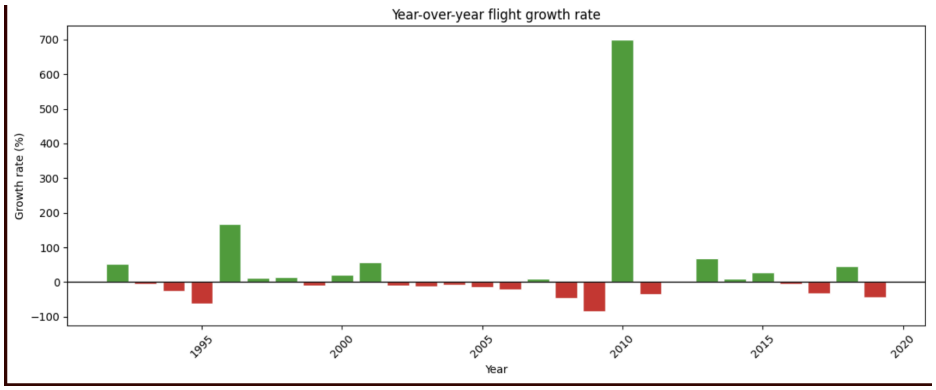
```python
if precip >= 1.0:
    return WEATHER_RAINY
```

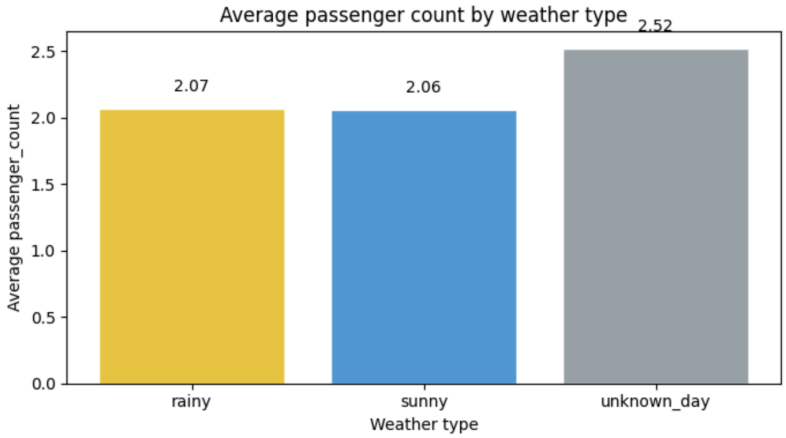
Visualizations:






Ratio: target destination / total
Lesley Groff
Nadia Marcinkova
Virginia Roberts Giuffre
Sarah Kellen
Larry Morrison
Brent Tindall
A P
Alan Dershowitz
Larry Visoski
Teala Davies
Ghislaine Maxwell
Emmy Tayler
Jeffrey Epstein
Jeff Epstein
Bill Hammond
Sophie Biddle
Doug Band
Bill Clinton
Andrea Mitrovich
Jean-Luc Brunel
Person
0
10
20
30
40
50
60
Percentage (%)


Year-over-year flight growth rate
700
600
500
400
300
200
100
0
−100
Growth rate (%)
1995
2000
2005
2010
2015
2020
Year

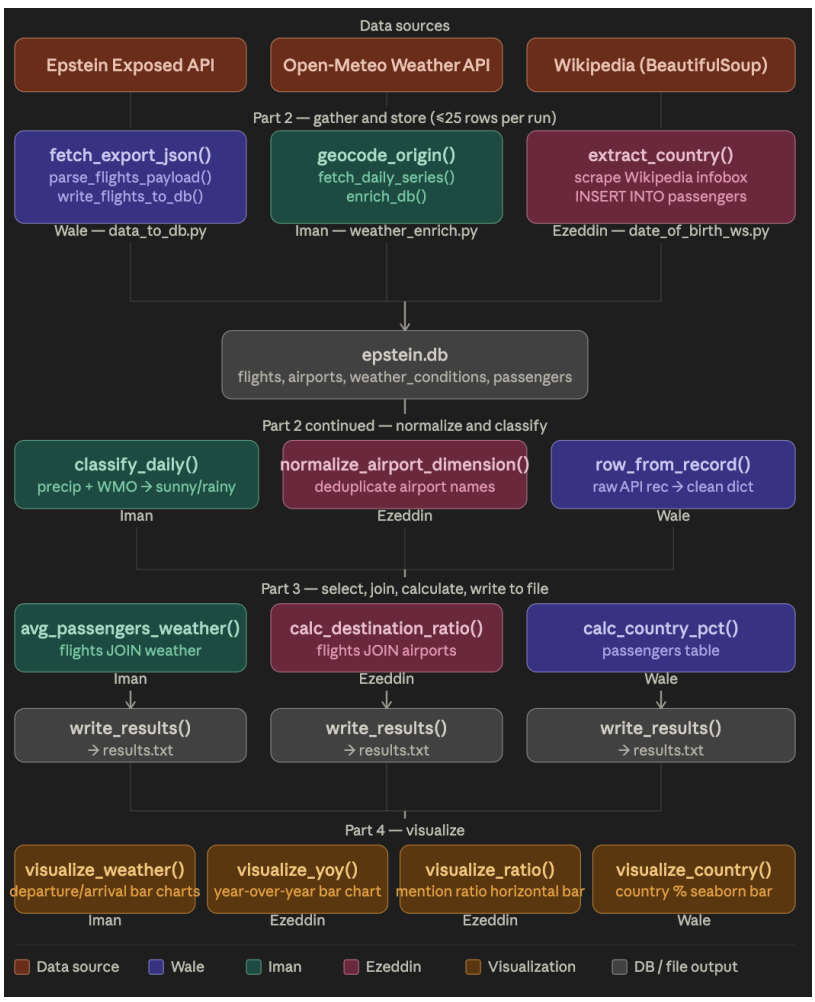

Average passenger count by weather type
2.52
2.07
2.06
2.5
2.0
1.5
1.0
0.5
0.0
Average passenger_count
rainy
sunny
unknown_day
Weather type

Instructions for running code:

Run in order:

1. data_to_db.py
2. Weather_enrich.py
3. date_of_birth_ws.py
4. calculation.ipynb
5. visualize_country_of_birth.ipynb
6. visualize_passenger_destination_ratio.ipynb
7. visualize_weather.ipynb

Updated Function Diagram:



Resource Log:

Date	Issue Description	Location of Resource	Result
April 20	Didn't know how to do a SQL JOIN	https://www.w3schools.com/sql/sql_join.asp	Helped solve it
April 22	Removing duplicate data in SQL database	https://learn.microsoft.com/en-us/troubleshoot/sql/database-engine/development/remove-duplicate-rows-sql-server-tab#:~:text=More%20information,table%20to%20a%20duplicate%20table.	Fixed duplicate data